Clean CIFAR-10 Baseline

Transfer-learning benchmark for temporal representation drift experiments

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87.07%	88.04%	9
FINAL TEST ACCURACY	BEST VALIDATION ACCURACY	SELECTED EPOCH

Executive summary

A clean transfer-learning baseline has been established for the semantic representation drift project. An ImageNet-pretrained ResNet-18 was used as a frozen feature extractor, while a compact 10-class head was trained on CIFAR-10. The selected checkpoint achieved **87.07% test accuracy** and is now the canonical no-drift reference for all later degradation and realignment comparisons.

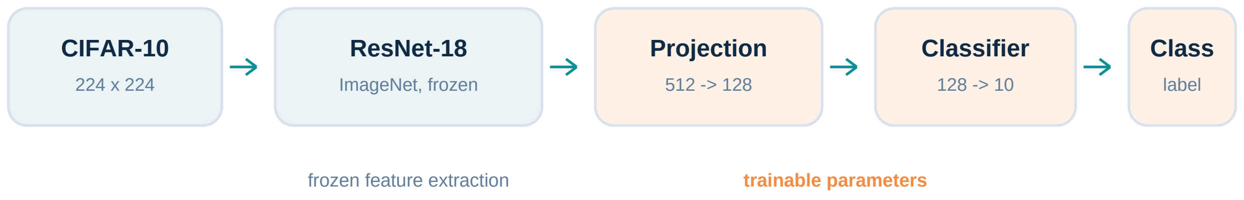
Completed baseline definition

Component	Canonical baseline choice
Dataset	CIFAR-10: 45,000 train / 5,000 validation / 10,000 test
Backbone	torchvision ResNet-18 with official ImageNet weights
Trainable head	512 -> 128 projection + ReLU -> 10-class linear classifier
Transfer policy	Frozen backbone and frozen BatchNorm running statistics
Input pipeline	224 x 224; ImageNet normalization; train-only crop and flip

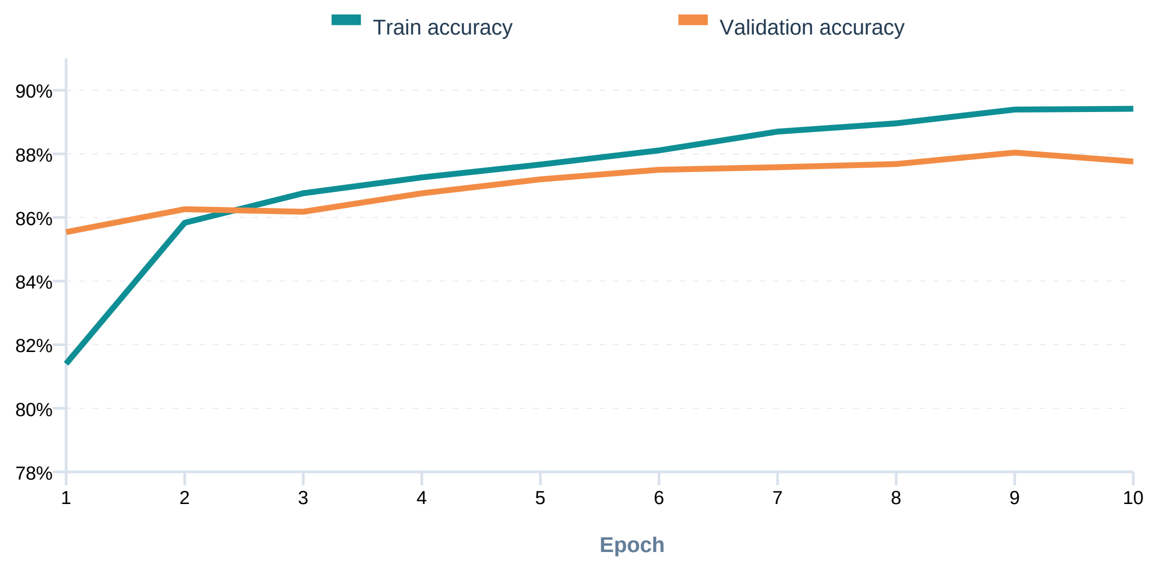
Baseline status: COMPLETE AND REPRODUCIBLE

Model and data flow

The pretrained convolutional backbone converts each image into a 512-D feature vector. Only the 128-D projection and the final 10-class classifier are optimized. This clean separation is useful later: the trained baseline can be wrapped as aligned transmitter and receiver endpoints without changing the reference task.



Training trajectory



Observed behavior

Validation accuracy improved from 85.54% to a peak of 88.04% at epoch 9. The final train-validation gap at the selected checkpoint was 1.35 percentage points, indicating stable head training with no large overfitting gap. Epoch 9 was selected before one-time test evaluation.

Exact training configuration

Parameter	Value	Parameter	Value
Optimizer	AdamW	Epochs	10
Learning rate	0.0010	Batch size	128
Weight decay	0.0005	Label smoothing	0.1
Scheduler	Cosine annealing	Seed	42
Checkpoint rule	Best validation accuracy	Device	Apple MPS

Per-epoch evidence

Epoch	Learning rate	Train loss	Train accuracy	Val loss	Val accuracy
1	0.001000	0.9858	81.40%	0.8856	85.54%
2	0.000976	0.8741	85.83%	0.8618	86.26%
3	0.000905	0.8491	86.76%	0.8551	86.18%
4	0.000794	0.8339	87.26%	0.8429	86.76%
5	0.000655	0.8265	87.66%	0.8308	87.20%
6	0.000500	0.8120	88.11%	0.8247	87.50%
7	0.000345	0.8032	88.70%	0.8212	87.58%
8	0.000206	0.7957	88.96%	0.8158	87.68%
9	0.000095	0.7886	89.39%	0.8139	88.04%
10	0.000024	0.7861	89.42%	0.8153	87.76%

Reproducibility and artifacts

The run is defined by **configs/baseline/cifar10.yaml**. Raw metrics, the resolved configuration, the best checkpoint, and the machine-readable summary are stored under **outputs/cifar10-resnet18-pretrained-baseline/**. The full run used seed 42 and completed in 1,066.14 seconds on Apple MPS. All five software tests and the code quality check passed after training.

NEXT MILESTONE	Wrap the baseline as an initially aligned transmitter-receiver pair and add a class-prototype semantic codebook.